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## Task 13 Report

For this spike we had to develop a scenario in which the Agents can behave as “prey” and “hunter”, allowing the hunter to wander around, and the prey to find tactical hiding spots running away from the hunter.

I kept the features of seeking, arrive, flee, wander and some debug visuals (With the keybind “i”).

I removed the features path following and pursuit, since they do not relate to the asked scenario.

I added the features hide, obstacles, hiding spots calculations and X markers on the debug feature for easier visuals.

### How it works:

The program runs a hide calculation for the prey, which “draws” a line from the hunter’s location to the center of an obstacle. That line then extends outside of the obstacle to create a hiding spot for the prey, this is to create a hiding point that’s between the obstacle and the hunter, blocking the sight for it.

After these hiding spots have been created, the prey chooses the one closest to its location, and with the “i” visuals, you can see how the chosen hiding spot turns green, as the others possible hiding spots are yellow. The hunter doesn’t really take in account neither the prey nor the obstacles and just wanders until it’s able to see the prey.

The hunter is Red while the prey is Green, the hunter has a bigger mass than the prey, lower max speed, less max force and it’s set on Wander mode by default, while the prey has Hide mode on default.